Microbiology Mid-Term 2023-2024

吕镇梅班期中卷，答案放最后了，小概率会出错~（整理 by 大鹿鼻窦）

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) Which of the following groups consist of organisms made up of prokaryotic cells?
A) protozoa and animals
B) Bacteria and fungi
C) Archaea and fungi
D) Bacteria and Archaea

2) Who is credited with using rRNA sequence comparisons to develop the three-domain system of life?
A) Robert Hooke
B) Frederick Griffith
C) Robert Koch
D) Carl Woese

3) Which of the following is NOT a property/process of all living cells?
A) transcription
B) differentiation
C) evolution
D) metabolism

4) Regarding early life on Earth, ____
A) microbial life existed for billions of years before plant and animal life.
B) microbial life, plant life, and animal life all appeared at about the same time.
C) microbial life existed long before animals but has been around for about the same amount of time as plants.
D) it is impossible to determine which type of life first appeared.

5) Transparent double-sided dishes used for growing microbes are most commonly called ____
A) baker dishes.
B) Petri dishes.
C) sterilization plates.
D) culture medium plates.

6) What was Pasteur's purpose in using a swan-necked flask rather than a traditional Erlenmeyer flask for his experiments on spontaneous generation?
A) to prevent air from entering
B) to be able to boil and sterilize the liquid medium
C) to prevent large debris and microbes from entering
D) all of the above

7) Which is the only group of microorganisms that contains lipopolysaccharides (LPS)?
A) gram-positive bacteria
B) *Bacteria*
C) gram-negative bacteria
D) Archaea

8) The cytoplasmic membrane is the ____
A) permeability barrier of the cell.
B) primary support structure of the cell.
C) source of nutrient production.
D) structure that identifies a cell as eukaryotic or prokaryotic.

9) The periplasm is a(n) ____
A) region between the cytoplasmic membrane and the outer membrane of gram-negative *Bacteria*.
B) part of the outer cell membrane of gram-negative organisms.
C) part of the inner cell membrane of gram-negative organisms.
D) alternate name for the inner cell membrane of any prokaryotic cell.

10) Which of these statements is true?
A) The gram-positive wall contains a thick peptidoglycan layer but is about as complex as the gram-negative wall.
B) The gram-positive wall contains a thick peptidoglycan layer but is less complex than the gram-negative wall.
C) The gram-positive wall contains a thick peptidoglycan layer and is more complex than the gram-negative wall.
D) The gram-negative wall contains a thick peptidoglycan layer and is more complex than the gram-positive wall.

11) Which statement is true regarding endospores and sporulation?
A) Sporulation is a type of reproduction and is an alternative to binary fission.
B) Calcium complexed with dipicolinic acid helps to dehydrate a developing endospore.
C) Endospores readily take up stain and thus are often seen as stained circles within vegetative cells.
D) The three steps of sporulation are activation, germination, and outgrowth.

12) Bacteria stain as gram positive or gram negative because of differences in the cell ____
A) wall.
B) cytoplasm.
C) nucleus.
D) chromosome.

13) Which of the following statements about transport mechanisms is incorrect?

A) ABC transporters consist of a substrate-binding protein, a transmembrane transporter, and an ATP-hydrolyzing protein.
B) Simple transport, group translocation, and ABC transport systems all require energy.
C) Proton motive force, ATP, or another energy-rich compound can be used to provide the energy required for active transport.
D) Active transport moves molecules from areas of high concentration to areas of low concentration.

14) The tetraether molecule within the membrane structure of the ____ yields a lipid____.
A) *Archaea* / monolayer
B) *Bacteria* / bilayer
C) *Archaea* / bilayer
D) *Bacteria* / monolayer

15) Which is NOT a function of capsules or slime layers?
A) evasion of destruction by immune cells
B) protection from osmotic lysis
C) attachment to solid surfaces
D) development and maintenance of biofilms

16) Chemolithotrophy involves ____
A) oxidation of organic compounds.
B) oxidation of inorganic compounds.
C) reduction of organic compounds.
D) metabolic autotrophy.

17) During fermentation, ATP is usually produced by ____
A) oxidative phosphorylation.
B) the citric acid cycle.
C) the proton motive force.
D) substrate-level phosphorylation.

18) If $\Delta G^{0\prime}$ is positive, ____
A) the reaction is exergonic and requires the input of energy.
B) the reaction is endergonic and requires the input of energy.
C) the reaction is endergonic and energy will be released.
D) the reaction is exergonic and energy will be released.

19) In aerobic respiration, when an organism uses lactose for energy, lactose acts as the ____ and oxygen acts as the ____.
A) Electron donor/electron donor
B) Electron donor/electron acceptor
C) Electron acceptor/electron donor

D) Electron acceptor/electron acceptor

20) In the process of cellular respiration, glucose is broken down into pyruvate via ____, and pyruvate is further oxidized to CO_2 through ____.
A) glycolysis / fermentation
B) the citric acid cycle (CAC) / glycolysis
C) Glycolysis / the citric acid cycle (CAC)
D) the citric acid cycle (CAC) / the glyoxylate cycle

21) One example of an electron acceptor that can be used in anaerobic respiration is ____.
A) Oxygen
B) Pyruvate
C) Nitrate
D) Water

22) In the context of a microorganism's metabolism, during glycolysis, which of the following is true?
A) the crucial product is CO_2; ATP is a waste product.
B) the crucial product is ATP; the fermentation products are waste products.
C) the crucial product is not relevant because glycolysis is not a major pathway.
D) the crucial product is ethanol or lactate; ATP is a waste product.

23) For microorganisms, growth is generally defined as an increase in ____.
A) the number of colonies present in a culture
B) the size of individual organisms
C) the number of cells
D) the life span of individual cells

24) The time interval required for the formation of two cells from one is called the ____.
A) growth rate
B) growth time
C) generation time
D) all of the above

25) The time between inoculation and the beginning of growth is usually called the ____.
A) log phase
B) lag phase
C) death phase
D) dormant phase

26) Under what conditions would a lag phase NOT occur?
A) when a stationary phase culture is transferred into fresh medium
B) when a culture subjected to UV light is transferred into fresh medium
C) when an exponentially growing culture is transferred into fresh medium

D) when a culture is transferred from a rich medium to minimal medium

27) Superoxide dismutase is indispensable to ____.
A) aerobes
B) aerotolerant anaerobes
C) obligate anaerobes
D) all of the above

28) Which of the following habitats is most likely to harbor an extremophile?
A) garden soil at neutral pH
B) permanently ice-covered lake
C) human skin
D) freshwater pond

29) A pure culture ____.
A) is sterile
B) is a population of identical cells
C) is made of a clearly defined chemical medium
D) was cultured for a certified stock culture

30) In a disc diffusion assay, the zone of inhibition will be ____ for organisms more resistant to an antimicrobial agent.
A) Absent
B) Smaller
C) Larger
D) yellow

31) Which of the following characteristics can apply to viral genomes?
A) Linear or circular
B) Double-stranded or single-stranded
C) Composed of RNA or DNA
D) All of the above

32) The packaging mechanism of T4 DNA involves cutting of DNA from ____.
A) circular genetic elements
B) linear genetic elements
C) DNA concatamers
D) none of these they are all transcribed directly from inserted viral DNA

33) A prophage replicates ____.
A) along with its host as long as the genes activating its lytic genes are not expressed
B) along with its host as long as the genes activating its lytic genes are being expressed
C) independently of its host as long as the genes activating its lytic genes are being expressed
D) independently of its host as long as the genes activating its lytic genes are not being expressed

34) A virus that kills its host is said to be ____.

A) virulent or lysogenic, but not temperate

B) Temperate

C) lysogenic

D) lytic or virulent

35) Which method is commonly employed to quantify bacteriophages?

A) Plaque assay

B) Spread-plate

C) Streak-plate

D) Spot titering

36) Which of the following statements about plaque assays is incorrect?

A) Viral titers are expressed as plaque-forming units (PFU).

B) Top agar containing a dilution of virions and bacteria is poured on an agar plate.

C) Plaques are clear zones seen because viruses lysed the host cells in the area.

D) Plaque assays are used to quantify lysogenic phages as well as lytic phages.

37) Which of the following accurately describes the process of T4 bacteriophage penetration into an *Escherichia coli* cell?

A) The envelope fuses with the cell membrane, allowing the phage nucleocapsid to enter the cell.

B) Integrase makes a pore in the cell wall, the tail sheath contracts, and the phage DNA is injected into the cell.

C) T4 lysozyme makes a pore in the cell wall, the tail sheath contracts, and the phage DNA is injected into the cell.

D) T4 lysozyme facilitates membrane fusion, allowing the phage to enter the cell.

38) Which of the following is true about *Archaea* and *Bacteria*, given that they lack a nucleus?

A) They use RNA polymerase for transcription.

B) They use polysomes to make multiple copies of a protein.

C) They supercoil their chromosomes.

D) They perform coupled transcription and translation.

39) Which of the following is a characteristic of RNA polymerase activity that is not true for DNA polymerase activity?

A) Precursor nucleotides can contain adenine, guanine, or cytosine.

B) Growth of nucleotide sequence occurs only in the 5'→3'direction.

C) Newly synthesized strands are antiparallel to their template strand.

D) A nucleotide primer is unnecessary.

40) When comparing genes from *Archaea*, *Bacteria*, and *Eukarva*, which statement is true?

A) Genes in *Archaea* involved in DNA replication, transcription, and translation are more similar to those of *Eukarya*, while those encoding metabolic functions other than information processing

are more similar to those of *Bacteria*.
B) Almost all archaeal genes are more similar to bacterial genes than to eukaryotic genes.
C) Archaeal genes are almost universally unique; very few are similar to either eukaryotic or bacterial genes.
D) Almost all archaeal genes are more similar to eukaryotic genes than to bacterial genes.

41) Housekeeping genes are present in ____.
A) chromosomes
B) plasmids
C) chromosomes and plasmids
D) neither chromosomes nor plasmids

42) Promoters are specific sequences of ____ that are recognized by ____.
A) RNA / DNA polymerase
B) RNA / RNA polymerase (coreenzyme)
C) DNA / RNA polymerase (coreenzyme)
D) DNA / sigma factor

43) Proteins interact predominantly within which portion of a double-stranded DNA helix?
A) major groove
B) minor groove
C) telomere
D) supercoil

44) Which of the following is incorrect regarding DNA and RNA synthesis?
A) DNA is the template for both DNA and RNA synthesis.
B) Both processes require an RNA primer to begin.
C) The template strand is antiparallel to the newly synthesized strand.
D) The overall direction of chain growth is from the 5' end to the 3' end.

45) Termination of RNA synthesis is ultimately determined by ____.
A) CG-rich sequences followed by AT-rich sequences.
B) specific nucleotide sequences on the DNA.
C) common termination sequences.
D) special protein factors.

46) AT-rich DNA will melt ____.
A) at a higher temperature than GC-rich DNA.
B) in accordance with the animal or plant from which it was taken.
C) at a lower temperature than GC-rich DNA.
D) usually at the same temperature as GC-rich DNA, with some minor variations.

47) The amino acid alanine is coded for by these codons GCU, GCC, GCG. This is an example of ____.

A) the wobble hypothesis.
B) the degeneracy of the genetic code.
C) the universality of the genetic code.
D) missense mutations.

48) Which of the following mechanisms allows pathogenic bacteria like *Escherichia coli* O157:H7 and *Staphylococcus aureus* to regulate the production of certain virulence factors until their population is sufficiently large?
A) Attenuation
B) Differentiation
C) Signal transduction
D) Quorum sensing

49) To repress transcription, corepressors bind to ____, and inducers bind to ____ proteins.
A) repressor proteins / inducer
B) repressor proteins / repressor
C) RNA polymerase / inducer
D) RNA polymerase / repressor

50) For the *lac* operon in *Escherichia coli*, under conditions in which glucose is absent and lactose is present, which of the following is true?
A) CRP and cAMP together are bound to the DNA, and LacI is bound to the operator site.
B) The lac operon is not transcribed because lactose is present.
C) CRP and cAMP together are bound to the DNA, and allolactose is bound to LacI.
D) The *lac* operon is not transcribed due to catabolite repression because glucose is absent.

Answer:

DDBAB	CCAAB	BADAB	BDBBC	CBCCB
CABBB	DCADA	DCDDA	ADABB	CBDBC